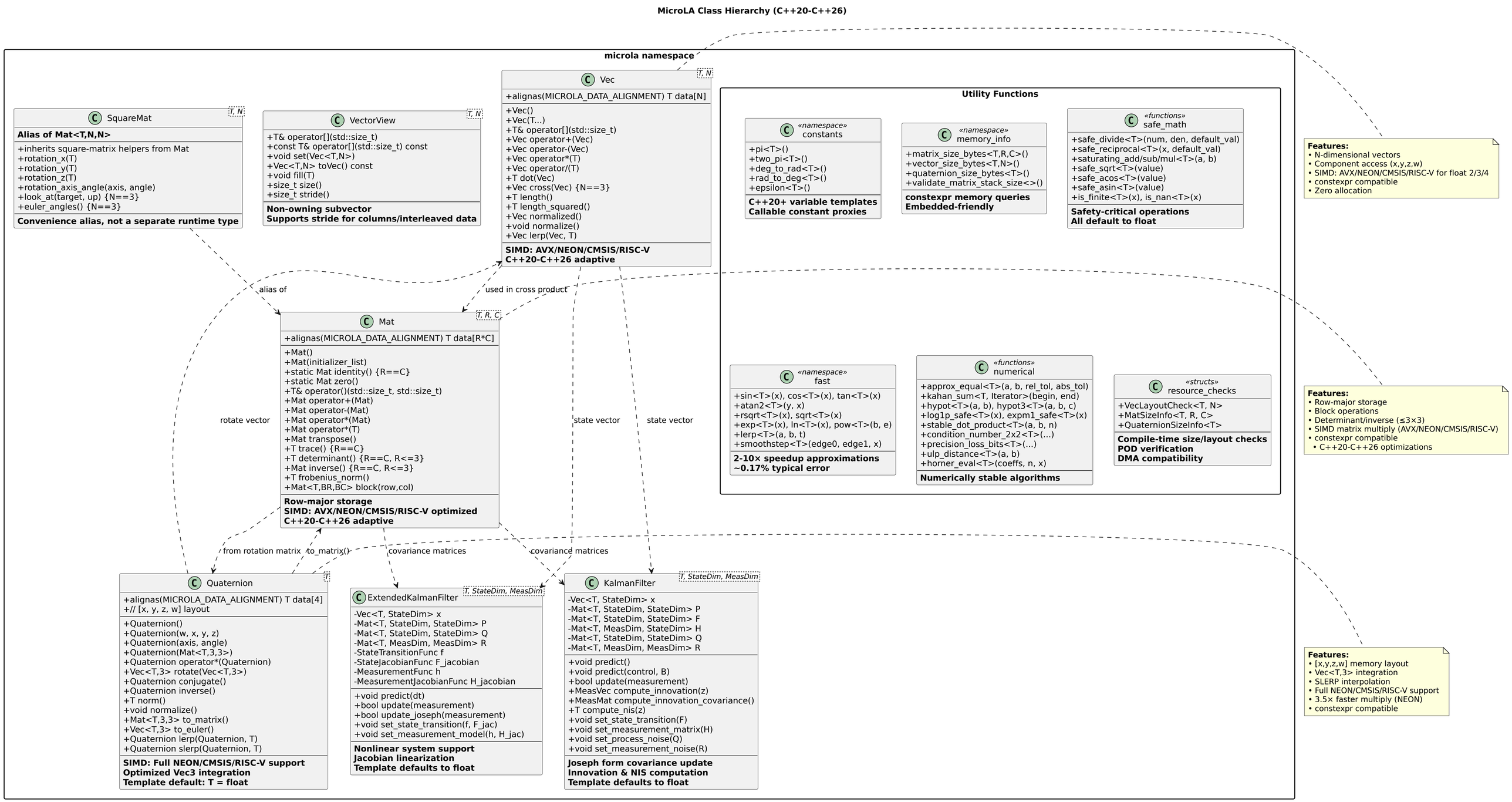
MicroLA Class Hierarchy (C++20-C++26)
microla namespace
Vec
T, N
+alignas(MICROLA_DATA_ALIGNMENT) T data[N]
+Vec()
+Vec(T...)
+T& operator[](std::size_t)
+Vec operator+(Vec)
+Vec operator-(Vec)
+Vec operator*(T)
+Vec operator/(T)
+T dot(Vec)
+Vec cross(Vec) {N==3}
+T length()
+T length_squared()
+Vec normalized()
+void normalize()
+Vec lerp(Vec, T)
SIMD: AVX/NEON/CMSIS/RISC-V
C++20-C++26 adaptive
SquareMat
T, N
Alias of Mat<T,N,N>
+inherits square-matrix helpers from Mat
+rotation_x(T)
+rotation_y(T)
+rotation_z(T)
+rotation_axis_angle(axis, angle)
+look_at(target, up) {N==3}
+euler_angles() {N==3}
Convenience alias, not a separate runtime type
VectorView
T, N
+T& operator[](std::size_t)
+const T& operator[](std::size_t) const
+void set(Vec<T,N>)
+Vec<T,N> toVec() const
+void fill(T)
+size_t size()
+size_t stride()
Non-owning subvector
Supports stride for columns/interleaved data
Utility Functions
«namespace»
constants
+pi<T>()
+two_pi<T>()
+deg_to_rad<T>()
+rad_to_deg<T>()
+epsilon<T>()
C++20+ variable templates
Callable constant proxies
«namespace»
memory_info
+matrix_size_bytes<T,R,C>()
+vector_size_bytes<T,N>()
+quaternion_size_bytes<T>()
+validate_matrix_stack_size<>()
constexpr memory queries
Embedded-friendly
«functions»
safe_math
+safe_divide<T>(num, den, default_val)
+safe_reciprocal<T>(x, default_val)
+saturating_add/sub/mul<T>(a, b)
+safe_sqrt<T>(value)
+safe_acos<T>(value)
+safe_asin<T>(value)
+is_finite<T>(x), is_nan<T>(x)
Safety-critical operations
All default to float
«namespace»
fast
+sin<T>(x), cos<T>(x), tan<T>(x)
+atan2<T>(y, x)
+rsqrt<T>(x), sqrt<T>(x)
+exp<T>(x), ln<T>(x), pow<T>(b, e)
+lerp<T>(a, b, t)
+smoothstep<T>(edge0, edge1, x)
2-10× speedup approximations
~0.17% typical error
«functions»
numerical
+approx_equal<T>(a, b, rel_tol, abs_tol)
+kahan_sum<T, Iterator>(begin, end)
+hypot<T>(a, b), hypot3<T>(a, b, c)
+log1p_safe<T>(x), expm1_safe<T>(x)
+stable_dot_product<T>(a, b, n)
+condition_number_2x2<T>(...)
+precision_loss_bits<T>(...)
+ulp_distance<T>(a, b)
+horner_eval<T>(coeffs, n, x)
Numerically stable algorithms
«structs»
resource_checks
+VecLayoutCheck<T, N>
+MatSizeInfo<T, R, C>
+QuaternionSizeInfo<T>
Compile-time size/layout checks
POD verification
DMA compatibility
Mat
T, R, C
+alignas(MICROLA_DATA_ALIGNMENT) T data[R*C]
+Mat()
+Mat(initializer_list)
+static Mat identity() {R==C}
+static Mat zero()
+T& operator()(std::size_t, std::size_t)
+Mat operator+(Mat)
+Mat operator-(Mat)
+Mat operator*(Mat)
+Mat operator*(T)
+Mat transpose()
+T trace() {R==C}
+T determinant() {R==C, R<=3}
+Mat inverse() {R==C, R<=3}
+T frobenius_norm()
+Mat<T,BR,BC> block(row,col)
Row-major storage
SIMD: AVX/NEON/CMSIS/RISC-V optimized
C++20-C++26 adaptive
Quaternion
T
+alignas(MICROLA_DATA_ALIGNMENT) T data[4]
+// [x, y, z, w] layout
+Quaternion()
+Quaternion(w, x, y, z)
+Quaternion(axis, angle)
+Quaternion(Mat<T,3,3>)
+Quaternion operator*(Quaternion)
+Vec<T,3> rotate(Vec<T,3>)
+Quaternion conjugate()
+Quaternion inverse()
+T norm()
+void normalize()
+Mat<T,3,3> to_matrix()
+Vec<T,3> to_euler()
+Quaternion lerp(Quaternion, T)
+Quaternion slerp(Quaternion, T)
SIMD: Full NEON/CMSIS/RISC-V support
Optimized Vec3 integration
Template default: T = float
ExtendedKalmanFilter
T, StateDim, MeasDim
-Vec<T, StateDim> x
-Mat<T, StateDim, StateDim> P
-Mat<T, StateDim, StateDim> Q
-Mat<T, MeasDim, MeasDim> R
-StateTransitionFunc f
-StateJacobianFunc F_jacobian
-MeasurementFunc h
-MeasurementJacobianFunc H_jacobian
+void predict(dt)
+bool update(measurement)
+bool update_joseph(measurement)
+void set_state_transition(f, F_jac)
+void set_measurement_model(h, H_jac)
Nonlinear system support
Jacobian linearization
Template defaults to float
KalmanFilter
T, StateDim, MeasDim
-Vec<T, StateDim> x
-Mat<T, StateDim, StateDim> P
-Mat<T, StateDim, StateDim> F
-Mat<T, MeasDim, StateDim> H
-Mat<T, StateDim, StateDim> Q
-Mat<T, MeasDim, MeasDim> R
+void predict()
+void predict(control, B)
+bool update(measurement)
+MeasVec compute_innovation(z)
+MeasMat compute_innovation_covariance()
+T compute_nis(z)
+void set_state_transition(F)
+void set_measurement_matrix(H)
+void set_process_noise(Q)
+void set_measurement_noise(R)
Joseph form covariance update
Innovation & NIS computation
Template defaults to float
alias of
used in cross product
rotate vector
state vector
state vector
from rotation matrix
to_matrix()
covariance matrices
covariance matrices
Features:
• N-dimensional vectors
• Component access (x,y,z,w)
• SIMD: AVX/NEON/CMSIS/RISC-V for float 2/3/4
• constexpr compatible
• Zero allocation
Features:
• Row-major storage
• Block operations
• Determinant/inverse (≤3×3)
• SIMD matrix multiply (AVX/NEON/CMSIS/RISC-V)
• constexpr compatible
• C++20-C++26 optimizations
Features:
• [x,y,z,w] memory layout
• Vec<T,3> integration
• SLERP interpolation
• Full NEON/CMSIS/RISC-V support
• 3.5× faster multiply (NEON)
• constexpr compatible